Jingoo Lee

Ph.D. Candidate · Civil, Urban, Earth & Environmental Engineering · UNIST

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EDUCATION

Ulsan National Institute of Science and Technology (UNIST) Ulsan, Korea

Combined Master's–Doctoral Program, Civil, Urban, Earth, and Environmental Engineering

Advisor: Prof. Young-Joo Lee

Mar. 2022 – Present · Expected Graduation: Feb. 2027

Ulsan National Institute of Science and Technology (UNIST) Ulsan, Korea

B.S., Mechanical Engineering

Mar. 2015 – Feb. 2022 (Military service: Jun. 2018 – May 2020)

RESEARCH INTERESTS

Deep learning for structural/seismic response prediction · Surrogate modeling for nuclear power plant (NPP) safety · Physics-informed and interpretable neural networks · Probabilistic uncertainty quantification · Virtual sensing and structural health monitoring · Fourier Neural Operators for PDE-based engineering problems

DOCTORAL DISSERTATION

Deep learning-based surrogate modeling for seismic response analysis of nuclear power plant structures

Doctoral Committee: Prof. Young-Joo Lee (Chair), Prof. Sukhoon Pyo, Prof. Myoungsu Shin, Prof. Junho Song, Dr. Minkyu Kim

Key Contributions

- Virtual sensing framework (Res-1D CNN) predicting seismic floor responses at 139 locations from a single seismometer.
- Attention-enhanced multimodal DL (Att-MMDL) for probabilistic seismic response prediction with structural uncertainty propagation.
- Physically interpretable 1D CNN: FIR filter–structural transfer function analogy enabling principled kernel size design.
- Windowed Recurrent Fourier Neural Operator (WR-FNO) for nonlinear seismic response prediction of multi-DOF systems.

JOURNAL PUBLICATIONS

Published

1. **Lee, J.**, Lee, S., Lee, Y.-J.*, and Lee, J.* (2025). Virtual sensing of seismic floor responses for rapid prioritization of critical equipment inspection in nuclear power plants, *Computer-Aided Civil and Infrastructure Engineering*, 40(26), 4669–4688.
2. **Lee, J.**, Lee, S., Lee, Y.-J.*, and Lee, J.* Predicting seismic floor response for nuclear power plant structures with timeseries uncertainty propagation using attention-enhanced multimodal deep learning, *Reliability Engineering and System Safety*.

Under Review

1. **Lee, J.**, Lee, Y.-J., Park, C.-S., Calmon, P., Miorelli, R., and Lee, J. Flaw characterization in steam generator tubes under deposit interference using eddy current testing via a Fourier Neural Operator, *NDT & E International*.
2. **Lee, J.**, Kim, M.*, and Lee, J.* Physically interpretable 1D CNN for seismic response prediction: A filter-theoretic approach to kernel design, *Nuclear Engineering and Design*.

In Preparation

1. **Lee, J.**, Lee, Y.-J.*, and Lee, J.* Recurrent Fourier Neural Operator for nonlinear seismic response prediction.

CONFERENCE PAPERS

International

1. Kim, M., Lee, S., **Lee, J.**, and Lee, Y.-J. (2023). Probabilistic prediction of structural response of cable bridges based on structural health monitoring data, *14th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP14)*, Jul. 9–13, Dublin, Ireland.
2. **Lee, J.**, Lee, S., and Lee, Y.-J. (2024). Seismic response prediction of auxiliary building using automated platform of sophisticated finite element analysis, *27th International Conference on Structural Mechanics in Reactor Technology (SMiRT 27)*, Yokohama, Japan.
3. **Lee, J.**, Lee, S., and Lee, Y.-J. (2024). Seismic response prediction of auxiliary building considering various structural properties and ground motions, *15th International Workshop on Advanced Smart Materials and Smart Structures Technology (ANCRiSST 2024)*, Jul. 10–11, Kyoto, Japan.
4. **Lee, J.** and Lee, Y.-J. (2024). Seismic response prediction of nuclear power plants considering uncertainty of structural properties and ground motions, *8th Asia-Pacific Symposium on Structural Reliability and Its Applications (APSSRA 2024)*, Sep. 29–Oct. 2, Ulsan, Korea.
5. Kim, D., **Lee, J.**, Lee, J., and Lee, Y.-J. (2025). Network capacity estimation based on centrality measures for building resilient road networks, *14th International Conference on Structural Safety and Reliability (ICOSSAR 2025)*, Jun. 1–6, University of Southern California, Los Angeles, CA, USA.
6. **Lee, J.** and Lee, Y.-J. (2025). Design of 1D CNN kernels for predicting seismic response of nuclear power plants, *28th International Conference on Structural Mechanics in Reactor Technology (SMiRT 28)*, Aug. 10–15, Toronto, Canada.

AWARDS & HONORS

Heki Shibata Early Career Award (Honorable Mention) Aug. 2025
SMiRT-28 International Scientific Committee · Toronto, Canada

RESEARCH EXPERIENCE

Graduate Researcher, Structural Reliability & AI Lab, UNIST Mar. 2022 – Present
Advisor: Prof. Young-Joo Lee

- Developed deep learning surrogate models (1D CNN, LSTM, Fourier Neural Operators) for seismic response prediction of NPP auxiliary buildings.
- Built virtual sensing framework estimating floor responses at 139 locations from a single seismometer.
- Designed probabilistic (attention-enhanced multimodal) DL for uncertainty propagation and real-time risk assessment.
- Derived physics-based kernel size selection rule using FIR filter theory and structural transfer functions.
- Collaborated with KRISS on eddy current testing and flaw characterization via Fourier Neural Operators.

TECHNICAL SKILLS

Programming: Python, MATLAB, Basic Linux, ABAQUS
Deep Learning: 1D CNN, Residual Networks, LSTM, Attention Mechanisms, Fourier Neural Operators
Structural Analysis: Finite element analysis, seismic response analysis, fragility assessment, Monte Carlo simulation
Signal Processing: FIR filter design, spectral analysis, sensor data processing

LANGUAGES

Korean (Native) | **English**